

Homework (Jalapeno)

- Q1.** The ancient Egyptians used $2x + 3y \geq 12$ to plan temple construction. Which inequality represents the same region?
- (A) $2x + 3y \leq 12$
 (B) $2x + 3y > 12$
 (C) $2x + 3y < 12$
 (D) $2x - 3y \geq 12$
 (E) $2x + 3y = 12$
- Q2.** The ancient Mesopotamians used linear inequalities to manage x sheep and y goats. If $x + 2y \leq 10$, what is the boundary line?
- (A) solid
 (B) dashed
 (C) $x + 2y = 10$
 (D) $x + 2y > 10$
 (E) $x + 2y < 10$
- Q3.** In ancient China, $3x - 2y \geq 6$ was used for taxation. Which region does this inequality represent?
- (A) above the line $3x - 2y = 6$
 (B) below the line $3x - 2y = 6$
 (C) on the line $3x - 2y = 6$
 (D) between $3x - 2y = 6$ and $x - y = 2$
 (E) left of the line $x - y = 2$
- Q4.** Ancient Greek architecture used $x + y \leq 8$ for column placement. What type of boundary line does this inequality have?
- (A) solid
 (B) dashed
 (C) $x + y = 8$
 (D) $x + y > 8$
 (E) $x + y < 8$
- Q5.** The Great Pyramid's base perimeter P was planned using $2x + 2y \geq P$. Which inequality represents the same region as $2x + 2y \geq 20$?
- (A) $x + y \geq 10$
 (B) $x + y \leq 10$
 (C) $2x + 2y \leq 20$
 (D) $x + y = 10$
 (E) $2x + 2y > 20$
- Q6.** Ancient Roman bridges used $x - 2y \leq 5$ for structural integrity. Which line is the boundary?
- (A) $x - 2y = 5$
 (B) $x - 2y > 5$
 (C) $x - 2y < 5$
 (D) $2x - y = 5$
 (E) $x + 2y = 5$
- Q7.** The Hanging Gardens of Babylon's irrigation used $y \geq \frac{1}{2}x + 2$. What type of line is the boundary?
- (A) solid
 (B) dashed
 (C) $y = \frac{1}{2}x + 2$
 (D) $y > \frac{1}{2}x + 2$
 (E) $y < \frac{1}{2}x + 2$
- Q8.** The Terracotta Army's layout was planned using $y \leq -x + 4$. What region does this inequality represent?
- (A) above the line $y = -x + 4$
 (B) below the line $y = -x + 4$
 (C) on the line $y = -x + 4$
 (D) between $y = -x + 4$ and $y = x - 2$
 (E) left of the line $y = x - 2$

Open Ended Questions (FRQ)

- Q9.** Graph the inequality $2x - 3y \leq 6$ and identify the boundary line. Explain your reasoning.
- Q10.** The ancient Mayans used $y \geq x - 3$ to plan their calendar. Graph this inequality and describe the solution region. Provide two points that satisfy the inequality.

Chef's Tips

1. Emphasize the importance of understanding boundary lines and their relationship to the solution region.
- Use historical contexts to make the problems more engaging and relevant.
- Ensure students can identify and explain the differences between solid and dashed boundary lines.